

Video Concepts

Learning Objectives

- Define video and explain how frames create motion
- Identify video formats and their uses
- Describe the difference between analog and digital video

Engage (5-7 minutes)

Think of a flipbook you made as a child—flipping pages quickly made drawings appear to move. Video works the same way: hundreds of still images (frames) shown every second create the illusion of motion. Today we learn how video works on computers.

Explore (10-12 minutes)

Watch a short video on the computer. Right-click on the video file and check its properties to find the format (MP4, AVI, MKV). Compare file sizes of two videos with different resolutions. Discuss which format is commonly used on phones and why.

Explain (10-12 minutes)

Video is a sequence of images called frames displayed rapidly to create motion. Common formats include MP4 (most compatible), AVI (large files), and MKV (high quality). Frame rate (fps) determines smoothness—24 fps is cinema standard, 30 fps is common for web video.

Elaborate (8-10 minutes)

Your cousin records village sports events on a phone. Which video format should he use for sharing on WhatsApp versus keeping for a school project? Write your recommendations with reasons, considering file size and quality. Compare analog TV signals with digital video streaming.

Evaluate (5-8 minutes)

Answer: What is a frame? Why is MP4 the most commonly used video format? What happens to video quality when you decrease the frame rate?